Pathfinder Project – Filtering, Clustering, and Structural Diversity in Enzyme Design

The proposed workflow extends traditional enzyme design pipelines by explicitly exploring the structural manifold of enzymatic proteins. Instead of focusing solely on reproducing known folds, the project aims to effectively discover structurally diverse enzyme mimics that preserve catalytic functionality while introducing novel scaffold architectures. Such diversity is essential for identifying alternative protein topologies capable of supporting the same catalytic mechanism, which may ultimately lead to improved stability, efficiency, or substrate specificity.

This exploration will be enabled through a computational pipeline that combines generative diffusion models, sequence design networks, structural prediction tools, and machine-learning–based analysis of protein structures. While generative models can produce a large number of candidate sequences and structures, experimental validation of catalytic activity remains costly and time-consuming. Consequently, it is essential to identify a small subset of candidates that are structurally diverse yet functionally plausible before laboratory testing.

Two complementary strategies will be used to ensure this diversity: filtering and clustering.

Filtering

Filtering acts as an initial quality control step that removes candidate enzymes that are unlikely to preserve catalytic functionality. In particular, candidate structures can be compared to the reference enzyme (e.g., turboPETase) using spatial matching of catalytic residues and structural similarity measures such as RMSD and sequence alignment. These filters ensure that the essential catalytic geometry is maintained, thereby preserving the biochemical feasibility of the designed enzyme.

However, structural prediction tools such as AlphaFold are known to exhibit internal biases toward common protein folds. As a result, simple filtering based solely on global similarity measures may still lead to a pool of candidates with limited structural diversity. To address this limitation, additional structural descriptors will be incorporated.

These descriptors include biologically meaningful features such as pocket volume, catalytic residue geometry, and predicted stability indicators. In addition, the protein structures will be represented as graphs, allowing the extraction of graph-topological descriptors (e.g., centrality measures, density, or graph radius). Certain combinatorial invariants, such as the size of minimum dominating sets, can provide information about structural compactness and interaction coverage within protein structures. Although computing these invariants is computationally challenging in general, efficient metaheuristic methods enable their approximation for large protein graphs within practical time limits. Collectively, these features provide a richer characterization of protein structures and enable more robust filtering and downstream analysis.

Clustering and consensus cluster testing

While filtering removes unsuitable candidates, clustering aims to organize the remaining structures into groups representing distinct regions of protein structure space. Biologically, such clusters correspond to families of proteins sharing similar structural organization or functional characteristics. Identifying these groups is important for ensuring that the final set of experimentally tested candidates covers multiple alternative scaffold architectures rather than many minor variations of the same fold.

Unsupervised machine learning methods may be used to cluster protein structures based on the extracted structural descriptors. Several clustering approaches will be explored, including spectral clustering, density-based clustering, and hierarchical clustering. Cluster quality will be evaluated using standard internal validation measures such as the Silhouette coefficient, which quantifies how well each structure fits within its assigned cluster.

Once clusters are identified, a consensus-based sampling strategy will be applied. Instead of selecting candidates from a single cluster, a portion of promising structures will be sampled from each cluster. This ensures that the final set of candidates represents multiple structurally distinct solutions. Structures not selected for experimental evaluation will be replaced by newly generated candidates from the generative model, allowing iterative expansion of the explored structural space.

Leveraging structural uncertainty

An additional source of diversification arises from the rich confidence information provided by AlphaFold. Metrics such as the predicted Local Distance Difference Test (pLDDT) score and the Predicted Aligned Error (PAE) matrix identify regions of structural uncertainty. These regions often correspond to flexible or poorly constrained structural elements. The workflow will preferentially target these uncertain regions for diversification, enabling exploration of alternative structural arrangements while maintaining the catalytic core.

Protein language model embeddings. Finally, the framework will incorporate representations derived from protein language models such as ESM-2, ProtT5, or ProGen. These models generate embedding representations of protein sequences that capture evolutionary, structural, and functional signals learned from large protein databases. Integrating such embeddings into the feature space is expected to improve clustering quality and facilitate the identification of structurally and functionally meaningful groups of candidate enzymes. Together, these filtering, clustering, and embedding-based strategies enable systematic exploration of the protein structural landscape while prioritizing candidates with high potential for catalytic activity. By combining generative models with advanced structural analysis, the project aims to uncover novel enzyme scaffolds that extend beyond the limits of naturally occurring protein folds.